THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Playing with Constructions** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** touch externally	Compare 17 with 11+6=17 and |11-6|=5.
2	**(a)** 20	Add the three sides.
3	**(c)** 36	Side $= 2r$.
4	**(a)** 16	Two diameters.
5	**(b)** 16	Each is half the diagonal.
6	**(a)** 18	Each side is $2r$.
7	**(b)** 70	Base angles are equal.
8	**(d)** 10	Diameter equals the side.
9	**(a)** touch internally	Compare 8 with 13+5=18 and |13-5|=8.
10	**(d)** 30	Add the three sides.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Intersect when $4 < d < 16$ (1); Here d = 12: intersect at two points (1).	1+1
12	$\angle AOB = \angle COD = 35^\circ$ (vertically opposite) (1); $\angle BOC = \angle AOD = 145^\circ$ (1).	1+1
13	Area of square $= 100$ cm^2 (1); Area of AOB $= 25$ cm^2 (one quarter) (1).	1+1
14	Intersect when $4 < d < 14$ (1); Here d = 15: do not touch (separate) (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Distance between adjacent centres $= 2r = 6$ cm (1); Side $= 6$ cm (1); Area $= 36$ cm^2 (1).	1+1+1
16	Longest chord = diameter $= 2 \times 9 = 18$ cm (1); Width of ring $= 5$ cm (1); Inner diameter $= 8$ cm (1).	1+1+1
17	BM $= 5$ cm, BN $= 3$ cm (1); Perimeter $= 2(5+3) = 16$ cm (1); Area $= 15$ cm^2 (1).	1+1+1
	OR: Distance between adjacent centres $= 2r = 8$ cm (1); Side $= 8$ cm (1); Area $= 64$ cm^2 (1).	1+1+1
18	Longest chord = diameter $= 2 \times 12 = 24$ cm (1); Width of ring $= 7$ cm (1); Inner diameter $= 10$ cm (1).	1+1+1
	OR: BM $= 7$ cm, BN $= 4$ cm (1); Perimeter $= 2(7+4) = 22$ cm (1); Area $= 28$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Draw AB = 6 cm (1); With A as centre and radius 10 cm draw an arc; mark the corner C using the perpendicular at B (1); Other side BC = 8 cm (1); Half diagonal OA $= 5$ cm (1); Perimeter $= 2(6+8) = 28$ cm (1).	1+1+1+1+1
	OR: Row length $= 5 \times 6 = 30$ cm (1); First to last centres $= (5-1) \times 6 = 24$ cm (1); Circles in the strip $= 7$ (1); Show the working with diameter (2).	1+1+1+2
20	Draw AB = 16 cm (1); With A as centre and radius 34 cm draw an arc; mark the corner C using the perpendicular at B (1); Other side BC = 30 cm (1); Half diagonal OA $= 17$ cm (1); Perimeter $= 2(16+30) = 92$ cm (1).	1+1+1+1+1
	OR: Row length $= 6 \times 8 = 48$ cm (1); First to last centres $= (6-1) \times 8 = 40$ cm (1); Circles in the strip $= 8$ (1); Show the working with diameter (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Playing with Constructions** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 100	Side $= 2r$.
2	**(c)** 24	Two diameters.
3	**(a)** 24	Each is half the diagonal.
4	**(a)** 30	Each side is $2r$.
5	**(d)** 55	Base angles are equal.
6	**(b)** 14	Diameter equals the side.
7	**(d)** one lies inside the other	Compare 3 with 12+7=19 and |12-7|=5.
8	**(d)** 32	Add the three sides.
9	**(d)** 196	Side $= 2r$.
10	**(c)** 32	Two diameters.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\angle AOB = \angle COD = 55^\circ$ (vertically opposite) (1); $\angle BOC = \angle AOD = 125^\circ$ (1).	1+1
12	Area of square $= 196$ cm^2 (1); Area of AOB $= 49$ cm^2 (one quarter) (1).	1+1
13	Intersect when $6 < d < 20$ (1); Here d = 8: intersect at two points (1).	1+1
14	$\angle AOB = \angle COD = 75^\circ$ (vertically opposite) (1); $\angle BOC = \angle AOD = 105^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Distance between adjacent centres $= 2r = 10$ cm (1); Side $= 10$ cm (1); Area $= 100$ cm^2 (1).	1+1+1
16	Longest chord = diameter $= 2 \times 10 = 20$ cm (1); Width of ring $= 4$ cm (1); Inner diameter $= 12$ cm (1).	1+1+1
17	BM $= 8$ cm, BN $= 5$ cm (1); Perimeter $= 2(8 + 5) = 26$ cm (1); Area $= 40$ cm^2 (1).	1+1+1
	OR: Distance between adjacent centres $= 2r = 12$ cm (1); Side $= 12$ cm (1); Area $= 144$ cm^2 (1).	1+1+1
18	Longest chord = diameter $= 2 \times 15 = 30$ cm (1); Width of ring $= 8$ cm (1); Inner diameter $= 14$ cm (1).	1+1+1
	OR: BM $= 9$ cm, BN $= 6$ cm (1); Perimeter $= 2(9 + 6) = 30$ cm (1); Area $= 54$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Draw AB = 10 cm (1); With A as centre and radius 26 cm draw an arc; mark the corner C using the perpendicular at B (1); Other side BC = 24 cm (1); Half diagonal OA $= 13$ cm (1); Perimeter $= 2(10 + 24) = 68$ cm (1).	1+1+1+1+1
	OR: Row length $= 7 \times 10 = 70$ cm (1); First to last centres $= (7 - 1) \times 10 = 60$ cm (1); Circles in the strip $= 9$ (1); Show the working with diameter (2).	1+1+1+2
20	Draw AB = 12 cm (1); With A as centre and radius 20 cm draw an arc; mark the corner C using the perpendicular at B (1); Other side BC = 16 cm (1); Half diagonal OA $= 10$ cm (1); Perimeter $= 2(12 + 16) = 56$ cm (1).	1+1+1+1+1
	OR: Row length $= 8 \times 4 = 32$ cm (1); First to last centres $= (8 - 1) \times 4 = 28$ cm (1); Circles in the strip $= 10$ (1); Show the working with diameter (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Playing with Constructions** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 36	Each is half the diagonal.
2	**(c)** 42	Each side is $2r$.
3	**(a)** 45	Base angles are equal.
4	**(a)** 18	Diameter equals the side.
5	**(a)** intersect at two points	Compare 17 with 9+9=18 and |9-9|=0.
6	**(a)** 36	Add the three sides.
7	**(d)** 400	Side $= 2r$.
8	**(d)** 36	Two diameters.
9	**(b)** 44	Each is half the diagonal.
10	**(d)** 48	Each side is $2r$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Area of square $= 324$ cm^2 (1); Area of AOB $= 81$ cm^2 (one quarter) (1).	1+1
12	Intersect when $0 < d < 16$ (1); Here d = 16: touch externally (1).	1+1
13	$\angle AOB = \angle COD = 100^\circ$ (vertically opposite) (1); $\angle BOC = \angle AOD = 80^\circ$ (1).	1+1
14	Area of square $= 484$ cm^2 (1); Area of AOB $= 121$ cm^2 (one quarter) (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Distance between adjacent centres $= 2r = 14$ cm (1); Side $= 14$ cm (1); Area $= 196$ cm^2 (1).	1+1+1
16	Longest chord = diameter $= 2 \times 13 = 26$ cm (1); Width of ring $= 5$ cm (1); Inner diameter $= 16$ cm (1).	1+1+1
17	BM $= 10$ cm, BN $= 7$ cm (1); Perimeter $= 2(10 + 7) = 34$ cm (1); Area $= 70$ cm^2 (1).	1+1+1
	OR: Distance between adjacent centres $= 2r = 16$ cm (1); Side $= 16$ cm (1); Area $= 256$ cm^2 (1).	1+1+1
18	Longest chord = diameter $= 2 \times 14 = 28$ cm (1); Width of ring $= 11$ cm (1); Inner diameter $= 6$ cm (1).	1+1+1
	OR: BM $= 6$ cm, BN $= 4$ cm (1); Perimeter $= 2(6 + 4) = 20$ cm (1); Area $= 24$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Draw AB = 8 cm (1); With A as centre and radius 10 cm draw an arc; mark the corner C using the perpendicular at B (1); Other side BC = 6 cm (1); Half diagonal OA $= 5$ cm (1); Perimeter $= 2(8 + 6) = 28$ cm (1).	1+1+1+1+1
	OR: Row length $= 9 \times 6 = 54$ cm (1); First to last centres $= (9 - 1) \times 6 = 48$ cm (1); Circles in the strip $= 11$ (1); Show the working with diameter (2).	1+1+1+2
20	Draw AB = 18 cm (1); With A as centre and radius 30 cm draw an arc; mark the corner C using the perpendicular at B (1); Other side BC = 24 cm (1); Half diagonal OA $= 15$ cm (1); Perimeter $= 2(18 + 24) = 84$ cm (1).	1+1+1+1+1
	OR: Row length $= 10 \times 8 = 80$ cm (1); First to last centres $= (10 - 1) \times 8 = 72$ cm (1); Circles in the strip $= 12$ (1); Show the working with diameter (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.